

Object Oriented Programming

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TASK 1

Description of Task 1:

Task 1 was to show yearly OHLC (Open, High, Low, Close) data for a chosen country in a table style. It was done in an organised and methodical way to make sure it was correct and easy to use. The job was to take weather data from different countries, figure out the yearly OHLC values for a certain country, and put the results in a table that was easy to read. This required parsing and organizing the temperature data, grouping it by year, and handling it efficiently while maintaining accuracy and clarity in the output.

Dealing with Data:

The CountryTemperature class was very important for keeping track of the weather information. The program was made to read and handle temperature data from a CSV file that had hourly temperature readings for several countries. The first row of the file was a header with the country numbers. The rows that followed had hourly weather readings for each country. A map structure (`std::map<std::string, std::vector<double>>`) was used to store these data points so that they would be easy to find and organise. The map's keys were country codes, and the corresponding values were vectors of hourly temperature readings.

The data was put into groups based on the idea that there are 365 days in a year, and each day has 24 hourly readings (365 * 24 entries). With this linked method, the temperature data for each country was split into yearly chunks, which made it possible to calculate the exact OHLC values.

Getting OHLC Values:

The OHLC numbers for each year were found by the `computeCandlestick` method in the CountryTemperature class. Here's how these numbers were found:

Open: The temperature at the beginning of the year or the Close number from the previous year is known as open.

High: This is the hottest time of the year, when the temperature reaches its highest point.

Low: This is the minimum temperature recorded, signifying the coldest point of the year.

Close: The average temperature over the year, calculated by summing all temperature readings and dividing by the total number of readings.

Using indexed calculations, this reasoning put data into groups by year, and the outputs were stored in Candlestick objects. It was easy to get the data and use it for more analysis or show because each object kept the year and its OHLC values.

User Interaction and Display:

The Menu class handled user interactions, ensuring a smooth and intuitive experience. A private method, `showCountryTemperature`, was responsible for generating and displaying the OHLC data for the selected country. This method used the `computeCandlestick` function to calculate the yearly statistics and displayed the results in a table format.

To ensure the table was visually appealing and easy to read, the `std::setw` function from `<iomanip>` was employed. This formatting technique aligned the columns for attributes like Year, Open, High, Low, and Close, providing a clean and professional appearance. The table header clearly labeled each column, and each row represented one year's data.

User Experience:

When the program started, users were presented with a menu displaying a list of available countries, each associated with a number. By entering the corresponding number, users could select a country and view its yearly OHLC data. If users made invalid entries, such as selecting a non-existent country, the program gracefully handled these errors by prompting the user to try again, ensuring robust input validation.

This design prioritized usability, ensuring users could navigate the system effortlessly while receiving accurate and organized information.

TASK 2

Description of Task 2:

The main goal was to show the OHLC data graphically as a candlestick graph, where each candlestick represents the annual temperature data for a certain country. The graph provides an intuitive way to observe trends and variations over the years.

How to Make a Candlestick Plot Graph:

1. Getting the Data Ready: The `computeCandlestick` method in the `CountryTemperature` class was used to get the candlestick data. Each candlestick contained:

Open: The weather at the beginning of the year (or the end of the previous year).

High: The hottest temperature that was recorded that year.

Low: The coolest temperature that was recorded all year.

Close: The average temperature for the year.

These OHLC values were formatted into a structure that could be processed by the candlestick plotting function.

2. Determining the Y-axis: The y-axis represents the temperature range, including a buffer for visualization clarity. The global high and global low temperatures were determined from the data, representing the highest and lowest temperatures across all years. A safety factor, like 20%, was used to make the range a little wider than the global high and low. This made sure that the candlesticks would fit nicely on the graph. The y-axis values were linearly spaced between the adjusted global low and high, with each step corresponding to a specific row in the graph.

3. Design for X-Axis: The years were shown on the x-axis. Each candlestick corresponded to one year. The years were labeled below the graph to align with the center of the respective candlestick. The candlestick width and spacing were adjusted for uniformity, ensuring the labels and candlesticks aligned properly. For a text-based graph, the years were displayed using substring format.

4. Drawing out the candlestick: Each candlestick visually represented the OHLC values.

Upper Stalk (High): This was a straight line that went from the highest Open/Close value to the High value.

Body (Open to Close): The Open and Close numbers were shown by a horizontal line.

The line was coloured:

Green: If $\text{Close} > \text{Open}$, it means that the direction is going up (warmer).

Red: If $\text{Open} > \text{Close}$, it means the direction is going down (cooling).

Lower Stalk (Low): This was a straight line that went from the bottom of Open/Close to the Low number.

To make this work in text: Loops were used to make a structure that looked like a grid, with a temperature level on the y-axis for each row. The candlesticks were drawn with ANSI escape codes for red and green and symbols (| for stalks and for bodies). Each year's candlestick was aligned with its position on the x-axis, ensuring consistency between labels and candlestick placements.

TASK 3

Description of Task 3:

It was supposed to let users: Choose a country's data from a choice and see details about that country. You can narrow down the data even more by choosing a number of years. Then, make a candlestick graph for those years. By focussing only on the important data, these limits were meant to make the system easier to use and more relevant.

Select Country Data Filter:

The country filter lets users pick one country from a list. Any actions taken after that will only affect the data from the chosen country, such as watching OHLC data or making candlestick graphs.

Menu Display: A list of countries was made by a method in the Menu class called "displayCountryList". The dataset's country codes were taken out and shown to the user as numbered choices, which made it easy to choose. User could type in a country code number or press 0 to go back to the main page.

Data Recovery: Once a country was picked, its code was sent to other methods, such as computeCandlestick or plotting methods, to make sure that only that country's data was handled.

Validation: Users couldn't choose incorrect options because of input validation. The program told the user to try again if they entered the wrong number.

Year Range Data Filter:

With the year-range selection, users could choose a beginning and finishing year. Then, the candlestick graph was made only for the data in this range, which made the study more focused.

Implementation Prompt for Year Range:After selecting a country, users were prompted to enter a starting year and an ending year.The range that was entered was checked to make sure it was correct. For example, the starting year could not be older than the finishing year, and both years had to be in the range of the available information.

Use of the filter:The CandlestickPlotter class's filterDataByYearRange method sorted the candlestick data by the range of years that were given.This method went through the candlestick data one entry at a time, comparing each entry's year to the range given, and only including entries that were in the range.

Graph Plotting: The data that had been sorted was sent to the candlestick plotting tool, which then plots the Candlestick graph.The graph used a clean and simple style to show the OHLC numbers for each year within the given range.

TASK 4

Description of Task 1:

The goal of Task 4 was to predict the OHLC (Open, High, Low, Close) values for the year 2020 using historical temperature data and to plot these predicted values using a text based graph. The 2020 OHCL values are computed by taking the average of the previous OHCL values. The assignment also asked for a reason for picking the model used for prediction.

1. Data Loading and Preparation

Description: The weather_data.csv file was used to load the past temperature data. The information was set up by country codes, and for each country, temperature readings were saved in vectors.

Implementation: The Country2020 class loaded and managed the temperature data.

The loadData method in Country2020.cpp read the CSV file, pulled out the country codes, and saved the weather data in a map where the country code was the key.

2. Predicting 2020 OHLC Values

Description: The OHLC values for 2020 were estimated by taking the average of the OHLC values that each country has had in the past.

Implementation: The compute2020 Values method in Country2020.cpp used past data to find the average Open, High, Low, and Close numbers.

To get the average, the method went through the candlestick data for each country, added up the OHLC values, and then divided by the number of years.

3. Plotting the Predicted OHLC Values

Description: The predicted OHLC values were visualized using a text based bar graph. Each bar represents one of the OHLC values (Open, High, Low, Close) for the year 2020.

Implementation: To make the bar graph, the plotBarGraph method in Menu.cpp was called.

The OHLC values were scaled to fit the height of the graph, and each value was shown by a different colour bar: green for "Open," red for "High," blue for "Low," and yellow for "Close."

4. Model Justification

Description: The chosen model for predicting 2020 OHLC values was based on the average of historical data. This approach was selected because:

Simplicity: Averaging is a straightforward and easy to understand method.

Stability: It provides a stable prediction by smoothing out anomalies in the data.

Applicability: Given the nature of temperature data, which tends to follow seasonal patterns, averaging provides a reasonable estimate for future values.

Limitations:

- The model does not account for sudden changes or trends in temperature data.
- It assumes that future temperatures will follow the same pattern as historical data.

ORIGINALITY

Data Filtering Approach:

The project introduced a dynamic data filtering system that allowed users to focus on specific subsets of data.

Country-Wise Filtering: Users could select a specific country from a list to isolate its data for analysis.

Year-Range Filtering: Users could further refine the data by specifying a range of years, ensuring the graph displayed only the most relevant entries.

This approach provided a highly interactive and customizable experience for users, making it possible to analyze specific data slices without overwhelming complexity. The use of a text-based candlestick visualization instead of graphical tools highlighted the lightweight and accessible nature of the solution, suitable for environments without graphical capabilities.

Use of ANSI Escape Codes:

The implementation of ANSI escape codes for color coding added an intuitive layer of interpretation to the text-based candlestick graph.

Green: Represented upward trends (Close > Open), signifying warming periods.

Red: Represented downward trends (Open > Close), indicating cooling periods.

This minimalistic yet effective use of color made it easier for users to identify trends at a glance while maintaining the lightweight, text-only nature of the solution.

Text based Graph for Predicted 2020 OHCL values:

To complement the data filtering capabilities, the project also implemented a text-based bar graph visualization for the predicted Open, High, Low, and Close (OHLC) values for the year 2020. Which is a simple but visually appealing and effective way to show the predicted OHCL data of 2020.

Dynamic Normalization: The bar graph dynamically normalizes the OHLC values based on the minimum and maximum values, ensuring that the representation is proportional and easy to interpret.

Color-Coded Representation: By employing ANSI escape codes, the graph visually distinguishes between the different OHLC values using color coding: green for Open, red for High, blue for Low, and yellow for Close. With this easy-to-understand colour scheme, users can quickly see trends and how the numbers relate to each other.

User -Friendly Output: The graph is displayed alongside the actual OHLC values, providing context and enhancing the user's understanding of the data. The structured layout, including y-axis labels and a clear x-axis, further contributes to the clarity of the visualization.

CHALLENGES

Dynamic Filtering:

Challenge: Implementing real-time filtering for both country selection and year range while maintaining efficiency and responsiveness.

Solution:

1. I organized data using efficient data structures like `std::map` and `std::vector` for quick lookups and filtering.
2. I precomputed candlestick data ensured seamless year range filtering without recalculating metrics.

Candlestick Representation:

Challenge: Designing proportional and interpretable text-based candlestick graphs particularly the stalks and bodies which required precision and iterative refinement.

Solution:

1. Stalks:
 - The upper stalk visually represented the range from the higher of Open or Close to the year's High temperature.
 - The lower stalk extended from the lower of Open or Close to the year's Low temperature.
 - These components captured the yearly temperature extremes effectively.
2. Bodies:
 - The horizontal segment between Open and Close, positioned within the stalks represented the yearly trend.
 - Color-coded bodies using ANSI codes added interpretive clarity to the otherwise text-based visualization.
3. Alignment:
 - Candlesticks were aligned with their respective year labels on the x-axis, ensuring clear correspondence between data and visual representation.

Encapsulating Multiple Tasks in the Menu:

Challenge: Designing a menu that combined the functionalities of the first three tasks into a single, cohesive option while maintaining usability was particularly difficult. The first option had to:

1. Display yearly OHLC values.
2. Plot the candlestick graph for a selected country and year range.
3. Seamlessly integrate filtering mechanisms for user inputs.
4. Ensure all these functionalities work together without overwhelming the user.

Solution:

The **Menu class** was carefully designed with clear separation of responsibilities with two options:

Option 1: encapsulated the first three tasks providing users with an end-to-end experience of selecting a country, applying year filters, and visualizing OHLC data through both tables and graphs.

Option 2: was reserved for displaying the predicted OHLC values for 2020. This separation ensured logical organization and simplified navigation for the user.

Clearly label all sections of the code that you personally wrote without assistance:

I have started the OOP midterm coursework from scratch and did not modify the folder provided. In the files I created, I marked my code with comments: `//my code` at the very top and bottom. However, in one

of my classes, CandlesticksPlotter, I received minimal assistance from my teacher. For this class, I added comments: //used my teacher's assistance in this code at the top and bottom.

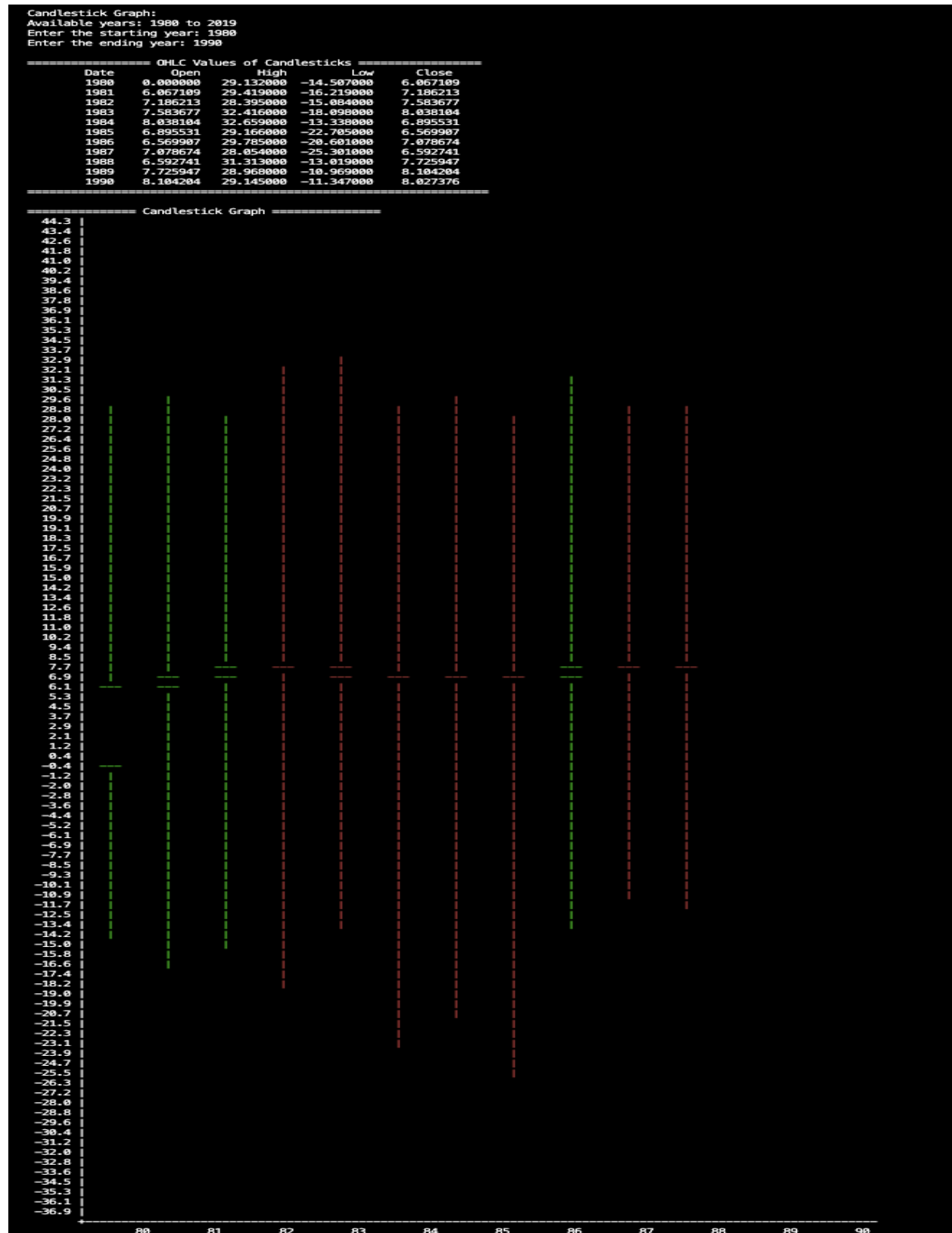
APPENDIX:

TASK-1 OUTPUT

```
===== Countries =====
1. AT
2. BE
3. BG
4. CH
5. CZ
6. DE
7. DK
8. EE
9. ES
10. FI
11. FR
12. GB
13. GR
14. HR
15. HU
16. IE
17. IT
18. LT
19. LU
20. LV
21. NL
22. NO
23. PL
24. PT
25. RO
26. SE
27. SI
28. SK
0. Back
Select a country by number : 1

Yearly OHLC Data for AT:
      Year      Open      High      Low      Close
1980          0      29.132     -14.507      6.06711
1981      6.06711      29.419     -16.219      7.18621
1982      7.18621      28.395     -15.084      7.58368
1983      7.58368      32.416     -18.098       8.0381
1984       8.0381      32.659     -13.338      6.89553
1985      6.89553      29.166     -22.705      6.56991
1986      6.56991      29.785     -20.601      7.07867
1987      7.07867      28.054     -25.301      6.59274
1988      6.59274      31.313     -13.019      7.72595
1989      7.72595      28.968     -10.969       8.1042
1990       8.1042      29.145     -11.347      8.02738
1991      8.02738      30.448     -15.978      6.96346
1992      6.96346      32.326     -13.679      8.12487
1993      8.12487      30.092     -17.096      7.33197
1994      7.33197      31.458     -12.935      8.64467
1995      8.64467       31.62     -13.081      7.45292
1996      7.45292      27.225     -17.757      6.09335
1997      6.09335      27.528     -20.283      7.13017
1998      7.13017      31.945     -15.665      7.84979
1999      7.84979      29.667     -16.172      7.79944
2000      7.79944      32.539     -16.425      8.77591
2001      8.77591      30.489     -17.771      7.69418
2002      7.69418      30.305     -16.596      8.35831
2003      8.35831      33.745     -17.103      8.13437
2004      8.13437      28.997     -15.015      7.31661
2005      7.31661      31.314     -19.122      7.08101
2006      7.08101       30.55     -21.521      7.50563
2007      7.50563      32.701     -10.583      8.61818
2008      8.61818      28.851     -11.306      8.41704
2009      8.41704      29.917     -15.502       7.7586
2010       7.7586      30.419     -18.188      6.89799
2011      6.89799      32.679     -13.038      8.34354
2012      8.34354       32.54     -17.832       8.2117
2013       8.2117      32.556     -15.969      7.63236
2014      7.63236      29.589      -8.895      9.17967
2015      9.17967      32.577     -11.174      8.93323
2016      8.93323      29.467     -12.179      8.49148
2017      8.49148      32.631     -14.708      8.30625
2018      8.30625      30.307     -17.596      9.20205
2019      9.20205      31.839     -10.777      9.21757
```

TASK-2 OUTPUT



TASK-3 OUTPUT

```
===== Menu =====
1. OCHL Values & Candlesticks
2. Predicting OCHL Values for 2020
0. Exit
Select one option: 1
===== Countries =====
1. AT
2. BE
3. BG
4. CH
5. CZ
6. DE
7. DK
8. EE
9. ES
10. FI
11. FR
12. GB
13. GR
14. HR
15. HU
16. IE
17. IT
18. LT
19. LU
20. LV
21. NL
22. NO
23. PL
24. PT
25. RO
26. SE
27. SI
28. SK
0. Back
Select a country by number : 1
```

Candlestick Graph:

Available years: 1980 to 2019

Enter the starting year: 1980

Enter the ending year: 1990

TASK-4 OUTPUT

```
===== Menu =====  
1. OCHL Values & Candlesticks  
2. Predicting OCHL Values for 2020  
0. Exit  
Select one option: 2  
===== Countries =====  
1. AT  
2. BE  
3. BG  
4. CH  
5. CZ  
6. DE  
7. DK  
8. EE  
9. ES  
10. FI  
11. FR  
12. GB  
13. GR  
14. HR  
15. HU  
16. IE  
17. IT  
18. LT  
19. LU  
20. LV  
21. NL  
22. NO  
23. PL  
24. PT  
25. RO  
26. SE  
27. SI  
28. SK  
0. Back  
Select a country by number : 2  
  
2020 OHLC Values (Computed Averages):  
Open: 9.5  
High: 29.7  
Low: -8.7  
Close: 9.7  
  
Predicted 2020 OHLC Values:  
Open: 9.5, High: 29.7, Low: -8.7, Close: 9.7  
Bar Graph:  
29.69 |||HHHHH||| ||| |||  
25.85 |||HHHHH||| ||| |||  
22.02 |||HHHHH||| ||| |||  
18.18 |||HHHHH||| ||| |||  
14.34 |||HHHHH||| ||| |||  
10.51 |||HHHHH||| ||| |||  
6.67 ||00000||HHHHH||| ||CCCCC||  
2.84 ||00000||HHHHH||| ||CCCCC||  
-1.00 ||00000||HHHHH||| ||CCCCC||  
-4.83 ||00000||HHHHH||| ||CCCCC||  
-8.67 ||00000||HHHHH||LLLLL||CCCCC||  
+-----+  
Open High Low Close
```